

Reforming European defence procurement to boost military innovation and startups

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Executive summary

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THE WAR IN UKRAINE has demonstrated how battlefields are evolving rapidly, with new technologies shaping military strategy and tactics. Increasingly, these new technologies are being provided by startup firms, rather than by established contractors. To incorporate these new firms into the defence ecosystem, European procurement practices need to be adapted. Global conflicts, including the United States/Israel war against Iran, further highlight the importance of strengthening Europe's defence industrial capacity.

WE SHOW THAT DEFENCE PROCUREMENT in selected European countries is primarily directed at the top-ten companies, with typically less than 30 percent of total order volume going to other firms. This contrasts with the United States, where the top-ten companies take up a substantially smaller share of the market. Israel and Ukraine also have robust policies in place to ensure the inclusion of young, innovative firms in military procurement.

EUROPEAN DEFENCE MINISTRIES must show greater willingness to enter into contractual arrangements with new players, as only contracts can lead to the necessary growth and scaling up of new technologies and the competition that is needed to reduce prices for defence products. Four steps should be taken:

1. Innovative firms need to be brought into the early phases of the defence acquisition process. The US Defense Innovation Unit and Germany's defence innovation centre are examples of how to facilitate the interaction between procurement offices and new firms.
2. Multiple companies should be involved in the development of innovative weapons technologies. While this means upfront development costs would be higher, competition in the deployment phase will reduce costs and increase technological leadership.
3. Requiring the inclusion of several companies of different sizes in framework contracts builds in competition and allows small players to grow.
4. Procurement decentralisation for small-scale and fast-developing technologies and public seed funding have been effective tools for accelerating technological innovation and giving fast access to new technology to commanding officers, who must otherwise wait for slow bureaucratic processes to play out.

Recommended citation

Kapstein, E., J. Ospital and G. Wolff (2026) 'Reforming European defence procurement to boost military innovation and startups', *Policy Brief* 04/2026, Bruegel, available at <http://doi.org/10.64153/DRNO4343>



1 Introduction

The experience in Ukraine of real-time battlefield innovation and rapidly changing defence technologies has generated a stream of lessons for military forces. The armament strategies of the United States, the United Kingdom and France, among others, are starting to reflect these lessons. The US Department of Defense¹, for example, issued in November 2025 an Acquisition Transformation Strategy to accelerate delivery of high-tech weaponry (DoW, 2025), while the UK Strategic Defence Review 2025 shows a substantive doctrinal shift characterised by the integration of conventional forces with digitally-enabled and increasingly autonomous systems such as drones (Ministry of Defence, 2025).

But is Europe's defence-industrial base ready to supply sufficient unmanned aerial systems and other weaponry of the sort deployed and tested in Ukraine?² Béraud Sudreau *et al* (2026) showed a substantial increase in the number of startup defence firms active in Europe, many specialising in new defence technology. But despite the emergence of these firms, actual procurement, especially within European Union nations, remains focused on established military systems and large incumbent defence manufacturers.

Transaction costs and information asymmetries tend to favour established procurement relationships. Consequently, a deliberate effort is needed to include new tech firms in the order process. Otherwise, defence procurement may miss significant innovation that exists outside the traditional defence industrial base, particularly by firms that do not primarily operate in military markets or by new firms. Ukraine has already successfully transformed its military procurement through a highly decentralised and digitalised system, sometimes described as an "*Amazon for weaponry*"³. Failing to benefit from such innovation would have implications for both security and growth (Gazzani *et al*, 2025).

A basic question for European governments updating their military strategies is therefore how procurement processes can be reformed to strengthen innovation and facilitate the entry of new firms. This goes beyond the scale economies and competition benefits of deeper defence market integration across European countries (as discussed by Mejino-Lopez and Wolff, 2025, and Wolff *et al*, 2025b).

This Policy Brief offers recommendations on the integration of new companies and emerging technologies into the defence procurement ecosystem. We argue that military acquisition officials can make changes to the way in which they buy weapons to encourage new entrants to bid for contracts, and that such changes are vitally important to EU countries as they pursue greater strategic autonomy. In section 2, we take stock of current procurement, based on the Kiel Military Procurement Tracker (Wolff *et al*, 2025a), and show that most contracts still go to large incumbent firms and to established technologies. In section 3, we analyse three US procurement reforms aimed at improving access for new firms. In section 4 we set out recommendations for European policymakers.

1 President Donald Trump in September 2025 signed an executive order rebranding the Department of Defense as the Department of War, but the legal name remains the Department of Defense. We use DoD throughout this paper.

2 Empirical evidence from open-source battlefield assessments in Ukraine indicates that unmanned aerial systems, including FPV (first-person view) drones and loitering munitions, have been responsible for approximately 70 percent to 80 percent of battlefield casualties and equipment losses since 2024. See Mark Boris Andrianijc, 'A Western-funded drone surge could end Russia's invasion of Ukraine', *UkraineAlert*, 14 July 2025, Atlantic Council, <https://www.atlanticcouncil.org/blogs/ukrainealert/a-western-funded-drone-surge-could-end-russias-invasion-of-ukraine/>; and Richard Thomas, 'Drones now account for up to 80% of casualties in the Ukraine-Russia war', *Army Technology*, 12 February, <https://www.army-technology.com/news/drones-now-account-for-80-of-casualties-in-ukraine-russia-war>. Analysts such as Bronk (2025) recommend that NATO should not aim to replace traditional firepower with drones, but rather to complement it, highlighting Russia's increasing effectiveness in intercepting and neutralising FPV uncrewed aerial systems and one-way attack drones.

3 Serhii Kuzan, 'Pick and Mix: Ukraine Opens an Amazon for Weaponry', *Insights & Analysis*, 3 September 2025, Center for European Policy Analysis, <https://cepa.org/article/pick-and-mix-ukraine-opens-an-amazon-for-weaponry/>.

2 European rearmament: slow and focused on old-tech and incumbent firms

It has proved difficult to address slow military procurement processes in Europe and the biases European procurement agencies have in favour of established players. Wambach *et al* (2023), in a report to Germany's Federal Ministry for Economic Affairs and Energy, emphasised the need to reform military procurement to enable experimentation and much faster acquisition. They argued that current procurement rules are ill-suited to rapid technological change and high-intensity conflict and that the Bundeswehr should integrate innovation at operational speed, rather than through lengthy, risk-averse programmes.

The report also called for greater military uptake of dual-use, civilian-derived technologies and private-sector R&D, allowing commercially-developed systems to be tested, adapted and deployed iteratively. In the past, similar concerns about delays and inefficiencies have prompted repeated discussions about reforming Germany's Federal Office of Bundeswehr Equipment, Information Technology and In-Service Support (BAAINBw, Bundesamt für Ausrüstung, Informationstechnik und Nutzung der Bundeswehr), the authority responsible for defence procurement⁴.

Germany's court of auditors, the Bundesrechnungshof, has reported years-long delays in getting from defence procurement to implemented capabilities. For example, sixteen years after commissioning the first K130 corvette for the German navy, five corvettes are in use, but there is still no capacity to fly drones from these vessels, though this is considered essential (Bundesrechnungshof, 2025). The example illustrates both slow procurement processes and the difficulty German procurement faces in integrating modern and by now standard technology into its rearmament efforts. Similar examples can be found in the US and most other European countries (for example, Kapstein and Oudo, 2009).

An analysis of procurement data from Wolff *et al* (2025a) shows that up to 2024, the lists of defence contractors in the United Kingdom, Germany and Poland contained only two start-ups. The UK contracted Anduril Industries for £17 million in 2023 to explore the implementation of its command-and-control software across the country's defence platforms⁵, and in 2021, Germany ordered four armoured vehicles as a proof of concept from Dutch start-up Defenture⁶. In 2025, German defence start-ups Quantum Systems, Helsing and Stark Defence received significant orders for drones⁷.

Figure 1 and Table 1 show how strongly defence procurement in Europe is geared towards large and established players. Figure 1 shows that the top ten contractors account for between 67 percent and 90 percent of military procurement in Germany, Poland and the UK. In the US, overall spending is much larger, making comparison difficult. The top 10 contractors in the US account for less than 40 percent of spending. A deeper look at how the US has changed its procurement strategy is therefore useful and we do this in section 3. Tables 1 and 2 show the

Up to 2024, the lists of defence contractors in the UK, Germany and Poland contained only two startups

4 See, for example, KPMG Case Study, 'German Ministry of Defence, Shaping the future of military procurement', January 2016, <https://assets.kpmg.com/content/dam/kpmg/pdf/2016/02/german-defense-case-study.pdf>.

5 UK government news of 2 November 2023, '£17million contract awarded for force protection technology', <https://www.gov.uk/government/news/17million-contract-awarded-for-force-protection-technology>.

6 Waldemar Geiger, 'Bundeswehr beauftragt Defenture mit der Entwicklung neuer Fahrzeuge für die Spezialkräfte', *ES&T*, 5 August 2021, <https://esut.de/2021/08/meldungen/ruestung2/29049/defenture-fahrzeuge-spezialkraefte/>.

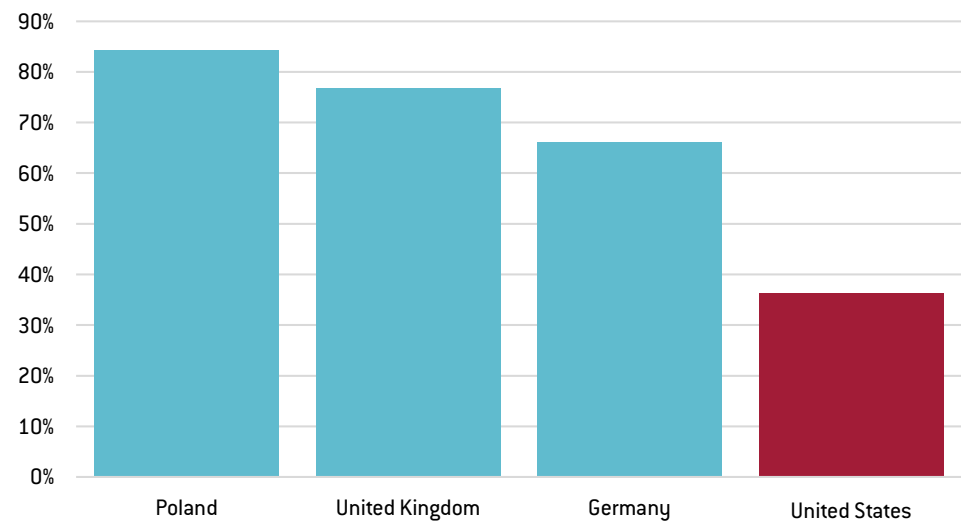
7 The German Bundeswehr ordered 520 Falke surveillance systems from Quantum Systems for €210 million, to be delivered in 2026; Michael Nienaber and Christina Kyriasoglou, 'Quantum Systems Set to Win \$246 Million German Army Drone Order', *Bloomberg*, 11 December 2025, <https://www.bloomberg.com/news/articles/2025-12-11/quantum-systems-set-to-win-210-million-german-army-drone-order>. Helsing and Stark Defence are set to receive contracts for €300 million each for the supply of drones; Laura Pitel, 'Daniel Ek and Peter Thiel-backed start-ups win German military drone contracts', *Financial Times*, 10 February 2026, <https://www.ft.com/content/eea27ca8-75e4-4d8a-90e7-bb98d90ece1>.

top ten defence contractors in Germany, Poland, the UK and US and the amounts going to these top contractors.

In any analysis, substantial differences in the defence industrial bases of the US and European countries should be taken into consideration. Figure 2 shows that US public spending on defence R&D is much higher than in European countries, both in absolute values and as a share of total public spending on R&D. The US provides public funding for basic research through the Defense Advanced Research Projects Agency (DARPA), which is tasked with developing emerging technologies for military applications, and which had a \$4.4 billion budget in 2025 (DARPA, 2025).

Meanwhile, while both the US and Europe have large numbers of smaller companies potentially available for major procurement, the US has 469 more defence start-ups than European countries (Figure 3). More broadly, going beyond defence, Europe (including Ukraine and the UK) has roughly one third of the start-ups of the US (EIB, 2019, 2020). European start-ups also attract less venture capital funding and are more likely than US start-ups to use public support.

Figure 1: Share of military procurement of the ten largest contractors in Germany, Poland, the UK and the US, 2020 to 2025



Source: Bruegel based on Wolff *et al* (2025a) and US General Services Administration.

Table 1: Top 10 defence contractors in Germany, Poland and the UK, € billions, 2024 prices

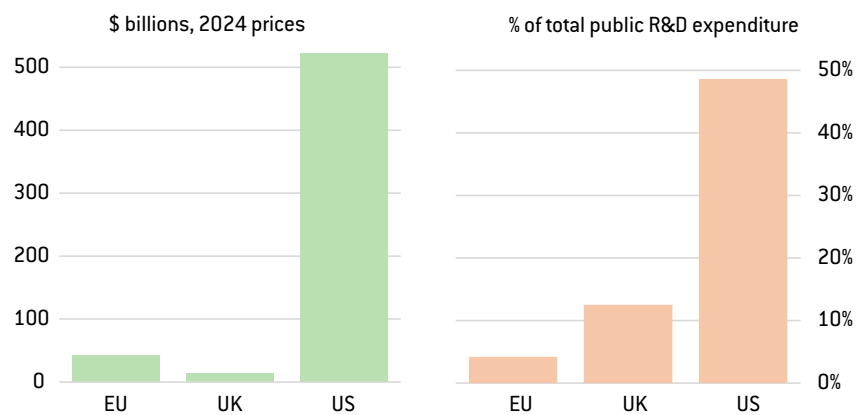
	2020	2021	2022	2023	2024	2025	Total
Germany	31.2	26.6	23.0	43.1	57.1	75.2	256.3
Bw Bekleidungsmanagement	0.3	0.1	10.1	0.0	3.0	20.5	34.1
Rheinmetall	0.8	1.7	1.0	2.3	8.5	12.7	27.0
Airbus	10.1	9.1	0.1	2.8	3.8	0.2	25.9
BWI	7.7	2.6	0.1	0.6	6.6	1.5	19.1
HIL GmbH	0.0	0.0	0.0	13.7	0.0	0.0	13.7
RTX	0.0	0.0	0.0	3.4	3.9	4.3	11.5
Damen Group	6.9	0.0	0.0	0.0	3.5	0.0	10.4
Boeing	0.0	1.3	0.4	8.5	0.0	0.0	10.2
Lockheed Martin	0.0	0.0	9.1	0.0	0.0	0.0	9.1
General Dynamics	0.0	0.0	0.0	0.0	0.0	8.5	8.5
All other companies	5.4	11.8	2.3	11.8	27.8	27.6	86.8
Equipment spending	13.2	11.9	10.5	13.2	19.8	n.a.	68.6
Poland	5.2	7.3	18.1	6.9	25.1	10.0	72.6
PGZ	0.2	2.3	2.1	0.6	6.2	3.3	14.7
Hanwha Group	0.0	0.0	6.2	2.3	1.5	0.0	9.9
Boeing	0.0	0.0	0.0	0.1	9.3	0.3	9.7
Lockheed Martin	4.9	0.0	0.0	0.0	1.6	0.0	6.5
General Dynamics	0.0	4.8	0.0	1.3	0.0	0.0	6.1
RTX	0.0	0.0	0.0	0.2	1.3	2.4	3.9
Hyundai Group	0.0	0.0	3.5	0.0	0.0	0.0	3.5
Korea Aerospace Industries	0.0	0.0	3.1	0.0	0.0	0.0	3.1
Leonardo	0.0	0.0	1.9	0.0	0.0	0.0	1.9
Kongsberg	0.0	0.0	0.0	1.8	0.0	0.0	1.8
All other companies	0.1	0.3	1.3	0.5	5.3	4.1	11.5
Equipment spending	4.7	5.4	4.6	10.9	13.7	22.0	61.3
United Kingdom	5.7	6.5	19.7	10.0	15.8	25.6	83.3
BAE Systems	3.3	0.9	11.9	6.1	0.1	0.4	22.6
Rolls-Royce	0.0	0.0	0.1	0.0	0.0	10.9	11.0
MBDA	0.0	0.8	0.4	0.0	8.2	0.6	9.9
Thales	0.7	0.6	0.0	0.2	2.6	1.6	5.7
Boeing	0.0	0.8	1.0	0.0	2.4	0.0	4.1
Lockheed Martin	0.3	0.3	0.2	0.3	0.0	1.4	2.5
Babcock	0.0	0.1	0.0	0.5	0.0	1.8	2.5
QinetiQ	0.4	0.0	0.0	0.0	0.0	1.7	2.2
BMT	0.0	0.0	0.0	1.9	0.0	0.0	1.9
Leonardo	0.4	0.0	0.6	0.1	0.0	0.4	1.6
All other companies	0.6	3.0	5.6	0.8	2.6	6.7	19.2
Equipment spending	18.0	22.5	20.4	22.0	25.9	29.7	138.5

Source: Bruegel based on Wolff *et al* (2025a) and NATO. Note: includes the 'main company' in procurement orders recorded in Wolff *et al* (2025a). Table A1 in the annex lists the companies included under 'All other companies'. 'Equipment spending' is based on NATO data, which provides estimates for 2024 and 2025.

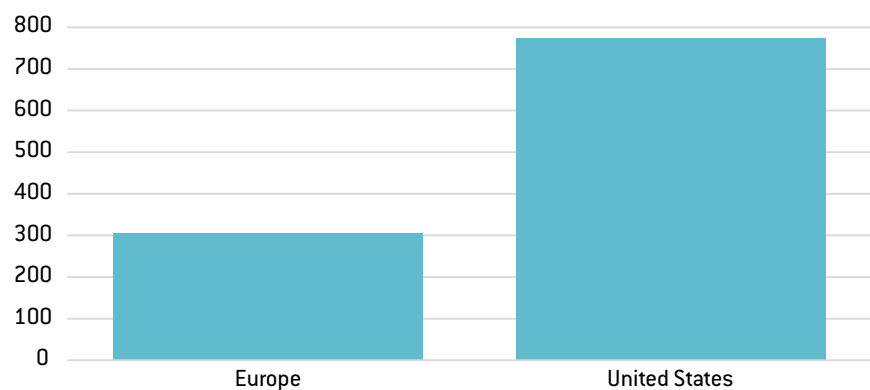
Table 2: Top 10 defence contractors in US procurement, \$ billions, 2024 prices

	2020	2021	2022	2023	2024	2025	Total
	699.2	508.8	409.0	456.9	445.9	450.3	2970.1
Lockheed Martin	124.0	52.6	45.2	68.7	41.5	48.9	380.9
RTX	23.9	26.9	25.4	30.0	24.3	30.0	160.4
Boeing	36.0	29.4	14.5	22.2	19.5	19.9	141.4
General Dynamics	36.4	22.7	21.3	22.6	3.4	0.7	107.0
Northrop Grumman	20.1	16.2	13.2	15.2	11.5	8.9	85.1
Huntington Ingalls	12.9	7.9	5.3	9.2	5.4	10.0	50.7
L3Harris	10.5	7.7	6.1	6.5	6.1	5.5	42.6
Humana	11.5	9.4	7.6	7.8	3.1	0.0	39.3
Pfizer	0.0	17.5	16.5	1.9	0.0	0.0	35.8
BAE Systems	10.8	4.8	5.0	7.5	2.5	0.0	30.5
All other companies	413.2	313.8	249.0	265.3	328.8	326.3	1896.4
Equipment spending	294.3	271.3	231.0	251.9	271.0	287.6	1607.1

Source: Bruegel based on US General Services Administration. Note: 'Equipment spending' is based on NATO data, which provides estimates for 2024 and 2025.

Figure 2: Public R&D spending on defence, 2020 to 2024

Source: Bruegel based on OECD.

Figure 3: Numbers of defence start-ups, as of January 2026

Source: Bruegel based on Preqin. Note: while most of the listed firms are defence start-ups, a small number of established or non-defence firms may have entered this data as well. We do not believe this changes the overall picture of a much larger number of defence start-ups in the US. Europe = EU, UK and Ukraine.

3 How US military procurement has changed to improve access for new firms

Procurement by the US Department of Defense (DoD) is a complex and lengthy process, and as such favours those firms that can navigate it. For example, planning for the F-35 fighter jet programme began during the Clinton administration, and resulted in an award to Lockheed Martin as the prime contractor. This process, governed by the Federal Acquisition Regulation (FAR), is primarily designed to acquire goods and services at scale: eventually, several thousand F-35s will be produced for US and allied forces.

There are three main stages in weapons procurement in the US (GAO, 2023; CRS, 2024):

1. Use of the Joint Capabilities Integration and Development System to identify requirements;
2. Running proposed acquisitions through the Planning, Programming, Budgeting, and Execution process (PPBE), which is used to allocate resources among competing programmes;
3. Mobilising the Defense Acquisition System (DAS) to develop and/or buy an item.

During this process, DoD officers are required to define quantities required, delivery schedules and evaluation criteria. Calls for tenders are then issued. Following technical and cost evaluations, procurement officers typically award contracts to single prime contractors⁸.

Mergers between US defence contractors during the 1990s increased the level of concentration and decreased competition in DoD procurement. Carril and Duggan (2020) found that these mergers led to an increase in contracts awarded without competition, though they did not find evidence of an increase in acquisition costs. NRC (2014) found that the standard US procurement process favoured firms with government contracting experience, because of the need to comply with extensive administrative and reporting requirements. In essence, DoD acquisition regulations were a barrier to entry to new entrants, especially small and innovative firms⁹.

In response, the DoD has implemented three approaches to encourage smaller and innovative firms to offer their goods and services.

1. The Small Business Innovation Research (SBIR) programme, overseen by the US Small Business Administration, funds research at small businesses. Launched in 1982, the programme was augmented in 1992 by the Small Business Technology Transfer (STTR) programme, which requires small businesses to partner with research institutions¹⁰. The DoD has made over 13,400 awards via SBIR/STTR during the programme's lifetime (NAS, 2026). In 2018, the US Air Force reformed its SBIR programme to introduce 'open topic' solicitations in which companies propose solutions to defence challenges.
2. Acquisition regulations allow procurement officials to set aside orders for small companies within multiple-award framework contracts. In 2013, reforms promoted the use of these framework contracts. Since this change in policy, multiple-award contracting has become the preferred contracting approach for small businesses, and the share of procurement from small businesses has increased.

8 Contracts may also be awarded without a competitive tender, particularly as follow-on awards to an existing contract, when the item is available from only one source or when competition would compromise sensitive information and pose a national security risk.

9 GAO (2025c) also found that this acquisition process leads to lengthy adoption timelines, with major defence acquisition programmes taking nearly twelve years on average to deliver a usable first version of a system.

10 The STTR requires awardees to work with organisations that have tech-transfer expertise (eg universities or federally funded research centres) to facilitate commercialisation. STTR was created to complement SBIR (GAO, 2025b). We include STTR within SBIR contracting.

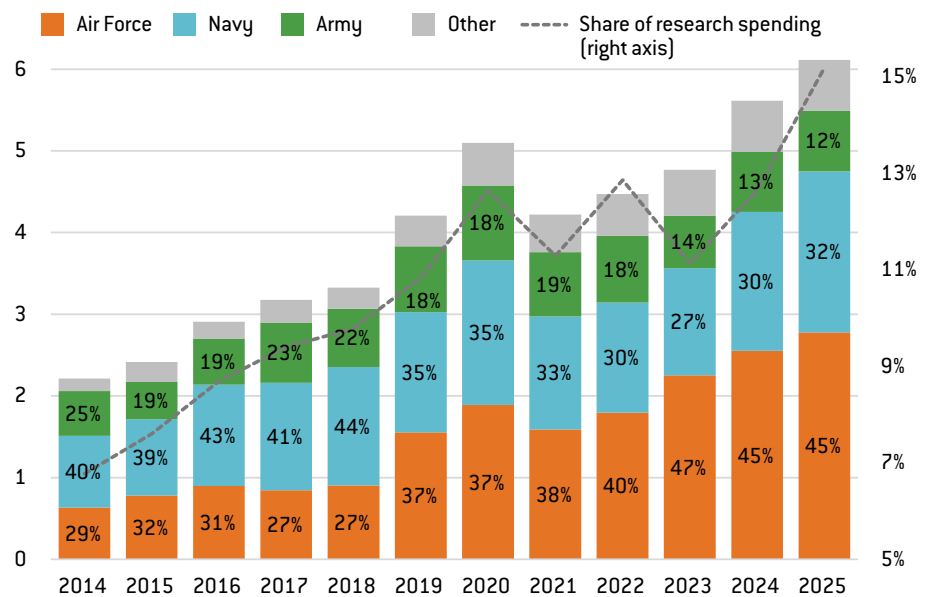
3. In 2015, the DoD created the Defense Innovation Unit (DIU) to fund the early-stage development of prototypes by startups. This organisation uses a bottom-up solicitation mechanism to introduce small and innovative firms into DoD procurement. Relatedly, the US Army has created an Executive Innovation Corps of reserve officers including senior Silicon Valley executives, helping to further integrate the firms into DoD processes and procedures.

In the following subsections, we detail these approaches and make a broad assessment of reform effectiveness. While this Policy Brief does not offer causal analyses, our figures and stylised facts indicate the effectiveness of these reforms in terms of boosting DoD procurement from small firms. In the context of the SBIR programme, Howell *et al* (2025) provide rigorous causal evidence on the effect of the 2018 US Air Force ‘open topic’ competition reform.

3.1 Small Business Innovation Research/Small Business Technology Transfer (SBIR/STTR)

The SBIR/STTR programme is the largest small business innovation programme in the US, allowing government departments to fund R&D in different technological areas¹¹. Of all government departments, the DoD uses the programme most. Since 2014, it has accounted for 80 percent of SBIR/STTR spending (see Figure A1 in the annex).

Figure 4: Value of DoD SBIR/STTR awards by agency, \$ billions, 2024 prices



Source: Bruegel based on US GSA. Note: ‘Share of research spending’ refers to SBIR contracting over total DoD spending on research. ‘Other’ refers to the remaining DoD agencies, including the Missile Defense Agency (MDA), US Special Operations Command (USSOCOM), the Defense Logistics Agency, the Defense Health Agency (DHA), the Defense Threat Reduction Agency (DTRA), and DARPA.

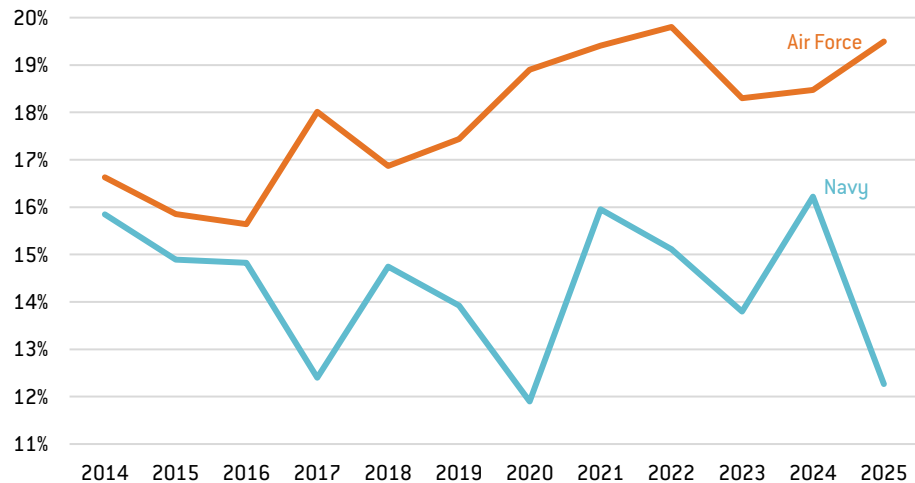
When the US Air Force reformed its SBIR programme in 2018 by introducing an ‘open topic’ solicitation mechanism (GAO, 2025b), the purpose was to reduce the contracting and administrative burdens that typically make DoD agencies unattractive customers for new entrants (GAO, 2025a). Since the 2018 reform, the DoD has increased its spending via this programme, both in total values and as a share of spending on research (Figure 4).

Since 2018, the DoD has awarded \$38 billion in SBIR/STTR contracts to small companies,

¹¹ The US Congress increased the funds set-aside for SBIR contracting through the 2011 SBIR/STTR Reauthorization Act, which required the DoD to allocate larger shares of its budget to SBIR and STTR (NRC, 2014).

amounting to 15 percent of total DoD research spending in 2025. From 2014 to 2018, among the DoD agencies, the navy was the largest user of this instrument. However, since 2019, the air force has become the largest user of the SBIR/STTR programme (Figure 4).

Figure 5: Share of total procurement going to small businesses at the Air Force and the Navy



Source: Bruegel based on US GSA data. Note: the share awarded to small businesses is calculated as the value of an agency's contracts awarded to small businesses over that agency's total contracts awarded.

Contracts awarded to small businesses by the air force, which introduced open-topic awards, now account for nearly 20 percent of its procurement. The navy's small business share is lower, declining to about 12 percent in 2025 (Figure 5). These trends are consistent with the findings of Howell *et al* (2021), who showed that winning open-topic SBIR awards increases small businesses' sales to the DoD, along with their numbers of patents and venture capital funding, while conventional SBIR awards have no significant effect on sales to the DoD. In its review of SBIR/STTR at the DoD, NAS (2026) found open-topic competitions have expanded participation in SBIR solicitations, and that companies that received SBIR/STTR awards account for roughly one third of all the firms conducting R&D for the DoD, though they receive just 11 percent of the department's spending on research.

3.2 Multiple-award contracting and small businesses

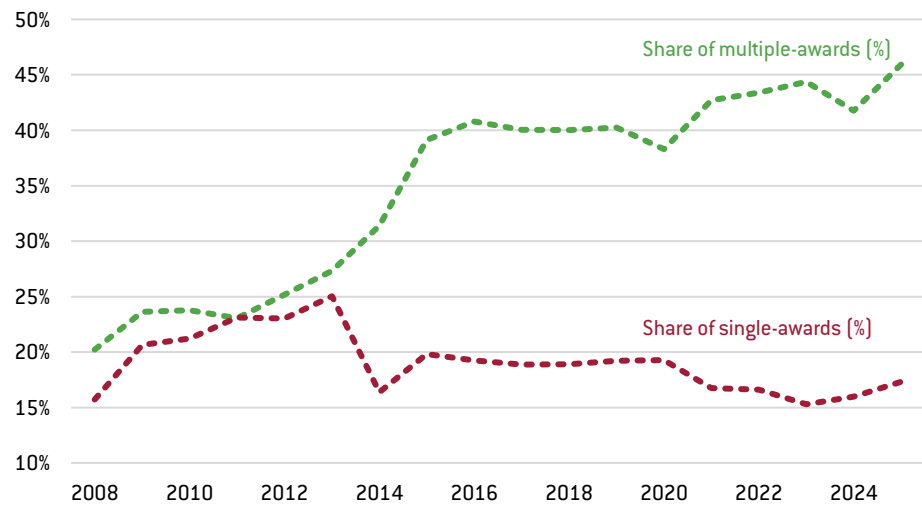
The DoD uses so-called indefinite-delivery, indefinite-quantity (IDIQ) framework contracts that can be used to purchase equipment or services without specifying quantities or timing of future orders at the time of the award (CRS, 2024). An IDIQ may be awarded either to a single company or to multiple vendors. For those awarded to multiple companies, vendors compete for subsequent orders under the same framework contract. Under these multiple-award IDIQs, procurement officers can reserve individual orders for small businesses (GAO, 2021). Since 2008, successive procurement reforms have promoted the use of multiple-award IDIQs, making them the preferred contracting method, and deterring the use of single-award IDIQs¹².

In particular, the 2013 Better Buying Power Initiative (BBP 2.0) directed procurement

¹² In 2008, the US Federal Acquisition Regulation (FAR) was reformed to make multiple award contracts the preferred contracting method, and to deter the use of single-award IDIQs by requiring procurement officials to obtain written approval from their agency heads for single award contracts exceeding \$100 million (Fusco, 2013). In 2020, the FAR was reformed again so that DoD officials were no longer required to consider price in awarding these framework contracts, making it easier for small companies to enter the pool of contractors that can submit proposals for an order under a multiple-contract award (CRS, 2024). Instead, prices are considered in the evaluation of proposals received for each order.

officials to select the type of contracting vehicle that would maximise competition throughout the life of a contract, promoting the choice of multiple-award IDIQ framework contracts¹³. These reforms seem to have increased small business opportunities (Figure 6)¹⁴.

Figure 6: Share of DoD awards to small businesses by contracting vehicle



Source: Bruegel based on US GSA. Note: 'share of multiple IDIQs' is the ratio of awards to small businesses over total multiple awards, while 'share of single IDIQs' is the ratio of awards to small businesses over total single awards.

Multiple-award IDIQ framework contracts have several features that benefit the inclusion of small businesses in DoD contracting. Because these contracts allow more than one vendor and do not require detailed technical or pricing evaluations at the time of award, they expand the pool of firms eligible to compete for orders (Fusco, 2013). Small businesses included in multiple-award IDIQs can expect larger future sales to the DoD than small businesses awarded contracts through single-award IDIQs (Table 3). After the 2013 implementation of BBP 2.0, the use of multiple-award IDIQs increased and the share of procurement awarded to small businesses rose from 17 percent to 20 percent in the following years (see also Figure A4 in the annex).

Table 3: Percentage change in small business sales to the DoD following inclusion in IDIQ framework contracts

	Central estimate	95% CI (low)*	95% CI (high)*
Multiple-award IDIQs	456%	298%	678%
Single-award IDIQs	168%	116%	233%

Source: Bruegel based on US GSA. Notes: percentage changes are calculated relative to small businesses that received other types of contracts (eg definitive delivery or definitive quantity contracts). Estimates are derived from log-point coefficients and converted to percentage changes. * The 95 percent confidence intervals show the highest and lowest values consistent with the data. Results show trends following the first inclusion of a small business in an IDIQ contract and should not be interpreted as causal effects.

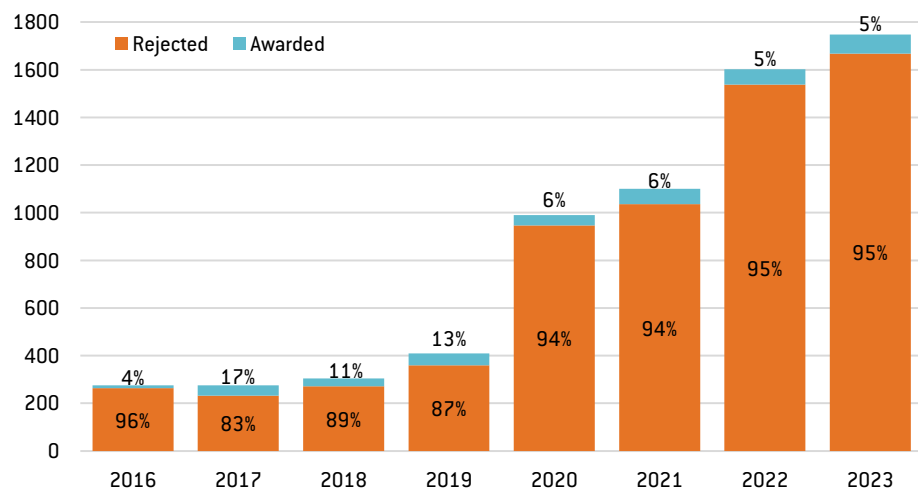
13 See Under Secretary of Defense, 'Implementation Directive for Better Buying Power 2.0 – Achieving Greater Efficiency and Productivity in Defense Spending', 24 April 2013, https://business.defense.gov/Portals/57/Documents/Attachment%209_BB%202%20Implementation%20Directive.pdf. The first BBP initiative, launched in 2010 by Under Secretary Ashton B. Carter, reformed acquisition processes by using contracting types to increase competition. The second BBP initiative focused on sustaining competition throughout the acquisition lifecycle and not just at the time of the award.

14 In 2025, orders from small businesses placed through multiple-award IDIQs amounted to \$37 billion, while small-business orders placed through single-award IDIQs were \$24 billion.

3.3 The Defense Innovation Unit (DIU)

In response to the slow take up of commercial technologies by DoD contractors, then Defense Secretary Ashton Carter established in 2015 the Defense Innovation Unit (DIU). Based in Silicon Valley, the DIU engages innovative companies in the US and allied countries to develop prototypes using commercially available technologies¹⁵. The DIU uses so-called commercial solutions opening (CSO) to call for proposals. CSOs set out a problem and invite companies to submit short proposals without stipulating technical requirements that restrict participation (GAO, 2025a). The DIU evaluates how viable proposals are and then awards contracts for prototype development¹⁶. Since 2017, prototypes that become operational products are published by the DIU in its Commercial Solutions Catalogue, for DoD agencies to procure¹⁷.

Figure 7: Number of proposals received by the DIU



Source: Bruegel based on DIU. Note: 'awarded' refers to 'other transaction agreements' (OTAs) granted to commercial companies, while 'rejected' means that a proposal was received but an OTA was not granted.

GAO (2025a) found that while the DIU has had limited success in scaling up solutions across the DoD, it has been increasingly successful in attracting proposals from startups. In 2020, the DIU received 1016 proposals in response to CSOs, more than double the previous year (Figure 7). Even though few of these proposals became operational products¹⁸, government reporting has found that the DIU has demonstrated the military application of commercial technologies¹⁹.

Since 2017, the DIU has published commercial solutions from 127 companies, 89 of which were small businesses. The US Defence Innovation Board (DIB, 2025) has recommended additional congressional funding so the DIU can expand its activities and focus on scaling up the production of commercial solutions. For example, the DIU has engaged with Anduril

15 DIU, 'DIU 3.0, Scaling Defense Innovation for Strategic Impact', 7 February 2024, <https://www.diu.mil/latest/diu-3-0-scaling-defense-innovation-for-strategic-impact>.

16 Contracts for prototype development are awarded through Other Transaction Agreements (OTAs), which allow the DoD to enter into research, prototyping and production contracts outside the normal procurement system (DIU, 2025).

17 See DIU, 'Commercial Solutions Catalog', <https://www.diu.mil/solutions/portfolio/catalog>.

18 Of all proposals submitted since 2016, about 6 percent have been awarded a contract for prototype development (Figure 7).

19 GAO (2025a) interviewed officials on six DIU projects that led to production contracts with the DoD. Officials highlighted frequent interactions between companies and the DoD and the lack of defined requirements in the DIU's acquisitions process as key drivers of the successful transition from prototypes to production. The successful projects reviewed in GAO (2025a) included an unmanned undersea vehicle, applications of AI to monitor disruptions in global navigation satellite systems and the use of commercial data to detect cyber threats.

Industries, a company created in 2017 that develops AI-enabled software for the autonomous control and command of defence platforms. Anduril also received SBIR/STTR awards in 2019, was included in a large multiple-award IDIQ contract in 2020 and by 2025 reached \$300 million in sales to the DoD²⁰.

4 Beyond the US: policies for defence innovation and small business in four countries

4.1 France

In 2018, France's defence ministry issued a broad policy framework to improve access for small and medium enterprises to defence procurement. This was updated in 2024 (Ministry of Armed Forces, 2024). It created an innovation agency (AID, *Agence de l'Innovation de Défense*) as an entry point for small businesses and startups. AID has multiple instruments to support innovation: grants, project-based funding and the opportunity to test prototypes with the armed forces²¹. In February 2026, AID announced a joint programme with Ukraine's defence innovation cluster to offer competitive grants for defence start-ups in both countries, and to allow the testing of equipment on the battleground²².

The small business policy also included reforms at the procurement agency (DGA, *Direction générale de l'armement*) to simplify acquisition procedures, reduce administrative burdens and shorten procurement timelines. DGA has placed orders with Exail, a French manufacturer of naval drones, awarded a €240 million contract to Prelingens for the AI processing of military data, and ordered 1000 tactical drones from the Paris-based startup Harmattan AI. According to the defence ministry, the share of defence spending reaching small and mid-sized increased from 14 percent in 2022 to 25 percent in 2024²³.

4.2 Germany

Germany inaugurated in February 2026 a new armed forces innovation centre (*Innovationszentrum Bundeswehr*) at a former airbase near Munich to address persistent weaknesses in military innovation and capability development. The initiative expands earlier pilot projects, such as the 'Innovation Lab System Soldier' (*Innovations Labor System Soldat*)²⁴ into a centrally-coordinated innovation hub. The centre's goal is to accelerate the innovation cycle by reducing gaps between concept development, testing and operational deployment.

20 DoD announcement of 23 September 2020, 'Contracts For Sep. 23, 2020', <https://www.war.gov/News/Contracts/Contract/Article/2358917/>.

21 Firms can receive support from the AID through grants for the development of dual-use technologies (RAPID), research (ASTRID and ASTRID Maturation), taking projects to mature defence technologies (PTD), projects to adapt commercial technologies (PAI) and participative or open innovation projects (AID, 2024).

22 AID and Brave 1, Ukraine's defence innovation cluster, have signed a letter of intent and the initiative is expected to start in the second quarter of 2026. See Ministry of Defence of Ukraine news of 25 February 2026, 'Strengthening defence cooperation with France: the new Brave France initiative', <https://mod.gov.ua/en/news/strengthening-defence-cooperation-with-france-the-new-brave-france-initiative>.

23 The 2024 figure includes startups. See Direction générale de l'armement press release of 18 June 2024, 'Le ministère des Armées renforce son plan d'action en faveur des PME, ETI et start-ups', <https://www.defense.gouv.fr/dga/actualites/ministere-armees-renforce-son-plan-daction-faveur-pme-eti-start-ups>.

24 For example, Heike Westhöfer, 'Bundeswehr erprobt im eigenem Innovationslabor Drohnen der nächsten Generation', Bundeswehr, 30 April 2025, <https://www.bundeswehr.de/de/organisation/ausrustung-baainbw/aktuelles/innovationslabor-drohnen-abwehr-5936478>.

Instead of the traditionally fragmented separation between research, procurement and end users, the centre will host soldiers, planners, researchers, procurement experts and industry partners, including startups and dual-use technology firms²⁵. However, the relationship between the centre and Germany's main military procurement authority, which is ultimately responsible for orders, is to be clarified. Industry has pointed out that an institutionalised innovation centre does not in itself guarantee procurement funds²⁶.

4.3 Ukraine

Following Russia's full-scale invasion in 2022, Ukraine's government amended the procedure for the supply of weapons to the armed forces. It issued a decree²⁷ that allowed use on the battlefield of new or experimental systems. Previously, a longer procedure was required before new equipment could be tested under battle conditions.

This regulatory change helped reduce the time taken for new systems to be deployed from one or two years to as little as two weeks (Bondar, 2025). Reforms at Ukraine's defence procurement agency have also allowed for the expedited procurement of drones. The government issues calls for drones directly via a digital platform; suppliers then submit applications and winners are determined through auctions²⁸.

In 2023, Ukraine published Brave1, a platform to provide financial and organisational assistance to defence technology projects²⁹. Applicants submit proposals to meet technological priorities aligned with battlefield needs; successful applicants are awarded grants ranging from \$11,000 for early-stage technologies to \$186,000 for the production and testing of products to be transferred to the armed forces³⁰. AI and missile technology proposals that represent a significant breakthrough can participate in a special programme for a grant of up to \$186,000, regardless of readiness levels. Brave1 also publishes competitions for grants of up to \$2 million for the development or scaling up of solutions that are frontline-ready. Solutions that develop into operational products can be transferred directly to the armed forces for battlefield use, and are added to a catalogue of items defence forces can order.

Over 1,500 companies have participated in Brave1 and, as of April 2025, it has provided 540 grants worth a total of \$50 million. To accelerate the deployment of these technologies, in April 2025, the government also launched the Brave1 Market, a platform via which army units can purchase defence technologies directly from manufacturers³¹.

4.4 Israel

Israel's defence ministry has a Directorate of Defense, Research and Development (DDR&D) that works to incorporate startups in defence procurement. According to the DDR&D, more than 300 startups have worked with the defence ministry, with orders to startups totalling about \$473 million since October 2023. Between October 2023 and September 2024, the defence ministry placed orders worth \$189 million with 86 startups and small companies, five

25 Ole Henckel, 'Pistorius eröffnet Innovationszentrum der Bundeswehr in Erding', Bundeswehr, 2 February 2026, <https://www.bundeswehr.de/de/meldungen/pistorius-innovationszentrum-erding-6064142>.

26 Oliver Rolofs, 'Innovationszentrum der Bundeswehr in Erding: Tempo für Innovation und Beschaffung', *ES&T*, 3 February 2026, <https://esut.de/2026/02/meldungen/67506/innovationszentrum-der-bundeswehr-in-erding/>.

27 Available at <https://ips.ligazakon.net/document/KP221097?an=1>.

28 Companies submit applications with cost assessment, valuations and proof of testing, answering calls from government units through the procurement platform Prozorro. See for example Government of Ukraine news of 22 May 2025, 'Ministry of Economy: State procures 50,000 drones for military via Prozorro Market', <https://www.kmu.gov.ua/en/news/minekonomiky-50-tysyach-droniv-dlia-viiska-zakupyla-derzhava-v-prozorro-market>.

29 See <https://brave1.gov.ua/en/>.

30 Zbroya Smart Force, 'Brave1 Grant Program', undated, <https://www.zbroya.gov.ua/en/services/brave1>.

31 This includes the purchase of drones, robots of various types, AI-based modules, electronic warfare and electronic warfare devices, and other components useful to the military in carrying out their missions. See Digital State UA, 'Ukraine Launches Brave1 Market for Defense Innovation', 28 April 2025, <https://digitalstate.gov.ua/news/tech/brave1-market-ukrayina-zapuskaye-marketpleys-viyskovykh-innovatsiy>.

times more than in the previous year³². The DDR&D incorporates these startups in defence procurement through four programmes:

The 'Green Lane' programme fast-tracks procurement from startups and small companies with annual revenues below \$7.8 million. Participants get simplified contractual conditions and assistance in navigating the bureaucratic processes involved in becoming a supplier to the defence ministry³³.

The Innotal programme identifies commercially available technologies with potential military applications and seeks to deploy them to the Israel Defense Forces (IDF) within six months. The programme evaluates the feasibility of proofs of concept through field evaluations and selects companies with the potential to meet the IDF's needs. Beneficiaries receive \$63,000 in initial funding, along with DDR&D guidance and the opportunity to conduct pilot tests with the IDF.

InnoFense is an accelerator programme for early-stage startups. It identifies companies in the early stages of exploring a technology and helps them develop technically feasible proofs of concept within six months. Companies are selected through regular open calls published by the DDR&D.

The Meimad programme offers academic institutions, companies and entrepreneurs funding for applied research with potential military and commercial applications³⁴. This programme is joint initiative with the Israel Innovation Authority and the finance ministry.

5 Reforming defence procurement to favour innovation

Defence procurement in EU countries can learn from the experiences of the US (section 3) and other countries (section 4) in terms of making the defence-industrial ecosystem more innovative. This is crucial because innovation and new technology need to play larger roles in military procurement to ensure that Europe's armed forces have access to the modern technology that is critical for modern warfare. It can also have positive spillovers to the wider economy.

Furthermore, in the current geopolitical circumstances, the share of European tech in European defence procurement should be increased to reduce one-sided dependence, for example for cutting-edge military software. European defence procurement remains highly dependent on US tech (Mejino-López and Wolff, 2025).

For defence innovation and defence startups, the problem of funding is no different to that faced by startups in other sectors (Philippon and Véron, 2008). However, new, high-tech defence companies need access to actual orders – this is the main factor in ensuring their growth. In this respect, what sets defence startups apart from other startups is the barriers to entry into procurement systems and the difficulty in eventually receiving substantial government orders.

Military practices are changing rapidly and lessons must be learned. Many of the new

32 Dean Shmuel Elmas, 'Jump in Defense Ministry orders from Israeli startups', *Globe*, 8 December 2024, <https://en.globes.co.il/en/article-defense-ministry-orders-from-israeli-startups-jumps-1001496278>; DDR&D, 'MAFAT For Startups', undated, <https://ddrd-mafat.mod.gov.il/en/mafai-for-startups>.

33 Via the Green Lane, suppliers benefit from a simplified pricing mechanism, the option to retain IP and a two-year exemption from ISO certification. Companies with up to ten employees also have their payments shortened to 30 days and are exempt from providing guarantees for contracts that do not involve physical deliveries. See DDR&D, 'Green Lane', undated, <https://ddrd-mafat.mod.gov.il/en/mafai-for-startups/green-lane>.

34 Depending on their project and type, organisations receive support ranging from \$170,000 to \$4.7 million, as well as assistance from DDR&D and the Israel Innovation Authority.

types of technology being used in Ukraine will require changes in procurement processes and different types of suppliers. To open military procurement to new firms, four main changes should be made to procurement policy.

1. *Bring innovative firms into the early phases of the defence acquisition process.* The DIU in the US (section 3.3) and Germany's new innovation centre (section 4.2) show how flexible government structures can be established to facilitate contacts between smaller, entrepreneurial firms and governments. However, while such relatively unbureaucratic agencies and initiatives may help to identify interesting new firms and products, this alone is not enough. Procurement agencies will still tend to prioritise large companies as less risky in fulfilling defence contracts. Small companies will be systematically disadvantaged as setting up contracts with them is similar to the effort needed for a big company, but big companies will bring larger volumes. Establishing innovation centres should thus eventually lead to contracts. This could be achieved, for example, by incentivising minimum levels of procurement from small and medium-sized companies³⁵. National legislators in Europe should consider mandatory minimum levels, as executive and procurement agencies do not have an incentive to make such changes.
2. *Multiple players should be involved in development of new tech.* All things considered, it is socially beneficial to have competition in the marketplace. Governments should fund several different companies for the development of specific new technologies. While this means higher costs in the development phase, development costs can be recovered later thanks to greater competition and lower prices for the final products. One example is the UK's Project Nightfall. It aims to develop a new ballistic missile for Ukraine, with the UK government selecting three "*industry teams*" to develop this technology, rather than giving an upfront contract to one firm. The goal is to get better technology at lower cost³⁶. Governments could make it a requirement that efforts to develop new technology should in general be supported at more than one company.
3. *Offer multiple vendor framework contracts.* New high-tech companies that pass the development phase will need to compete for orders. Governments should give orders to multiple companies whenever possible, via framework contracts. Multiple companies offering supplies can generate greater production capacity, as firms have to compete on the basis of credible offers to produce by deadlines. Variety in manufacturing could also contribute to military resilience by providing insurance against attacks against any one facility, or site-specific production problems. Multiple companies in a framework contract also ensure competition and greater price effectiveness of procurement. Bureaucratic practices could be changed by introducing mandatory multiple framework contracts, at least for less-specialised products.
4. *Increase procurement decentralisation for small-scale and fast-developing technologies.* Ukraine offers important lessons for procurement reform under battlefield conditions. While some of the reforms Ukraine has implemented may seem too audacious for armies outside of battle, national policymakers should lay the foundations by building infrastructures for more decentralised procurement through a government-established marketplace. More decentralised procurement would create a double benefit for military commanders and for the ecosystem of defence companies. For example, Germany has not yet been able to find ways of procuring drones for its K130 corvette. Empowering and

35 The US in fact has put such a target in place (GAO, 2021), whereby a certain share of procurement values is directed at small businesses. However, the US administration has said this will be reviewed. See Chris Gonzalez-Tablada, '8(a) Client Alert: Secretary Hegseth Directs Line-by-Line Review of All Small Business Sole-Source and Set-Aside Awards Over \$20 Million', Maynard Nexsen, 23 January 2026, <https://www.maynardnexsen.com/publication-8-a-client-alert-secretary-hegseth-directs-line-by-line-review-of-all-small-business-sole-source-and-set-aside-awards-over-20-million-1>.

36 See UK government press release of 11 January 2026, 'UK to develop new deep strike ballistic missile for Ukraine', <https://www.gov.uk/government/news/uk-to-develop-new-deep-strike-ballistic-missile-for-ukraine>.

providing limited budgets to military leaders, such as captains of ships, to order directly from a marketplace, may be a way to foster military innovation and fix holes in military capacities that current procurement agencies and processes have been unable to fix.

The European market for defence products is fragmented, with national procurement agencies and national markets. This means some markets may be too small to support the emergence of a new defence ecosystem. Fragmentation may also prevent the scaling up of companies into players that can innovate and produce at competitive prices. Deeper defence market integration, for example in the context of providing military support to Ukraine (Kirkegaard, 2025), or more ambitiously at a European scale (Wolff *et al*, 2025b), are critically important, in particular at a time when the transatlantic defence relationship is subject to major political stress.

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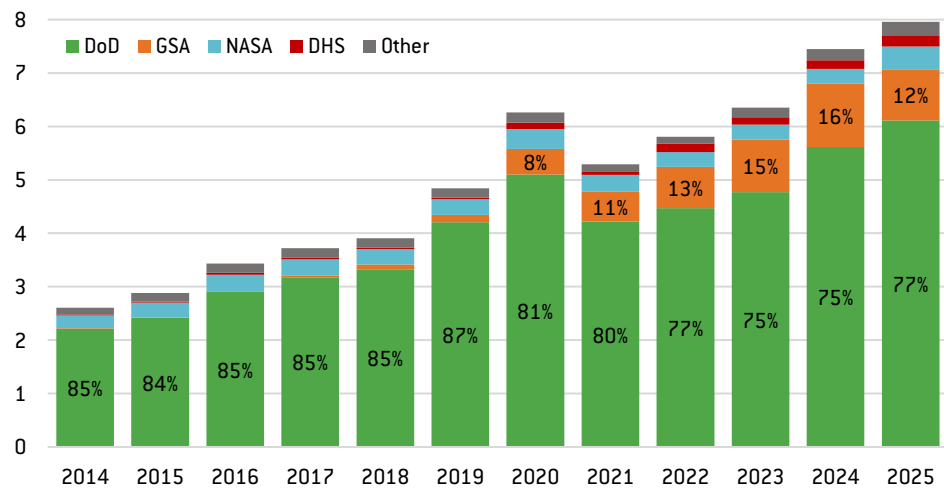
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Annex

Table A1: European companies included in 'all other companies' in Table 1

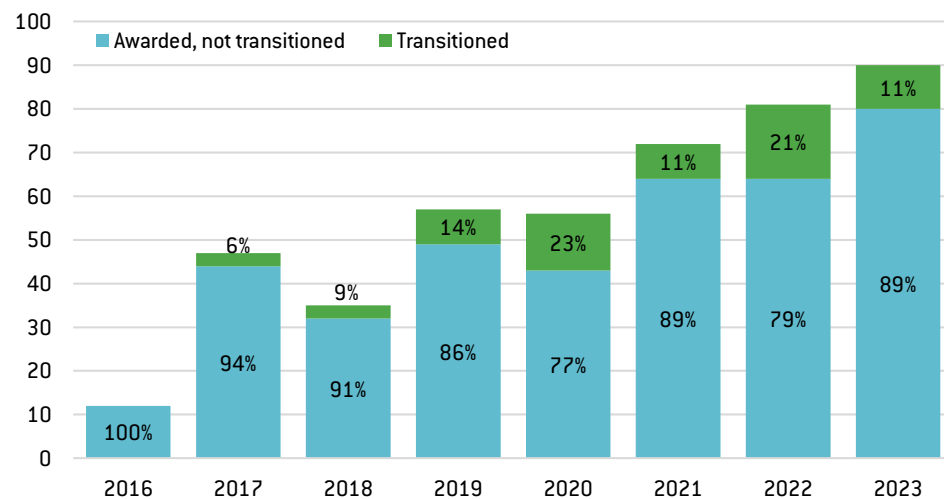
Germany		United Kingdom		Poland
Hensoldt	Fassmer	Rheinmetall	CGI	Kongsberg
KNDS	GIWS	Foreland Shipping	Marine Specialised Technology	Saab
Israel Aerospace Industries	EuroSpike	RTX	Terberg Group	Fabryka Broni 'Łucznik'
NATO	Dynamit Nobel Defence	Google	Team Protect	Remontowa Holding
General Dynamics	Motorola Solutions	US Department of Defense	Esri	General Atomics
Lürssen	Steep	Airbus	HeliOperations	Baykar
MBDA	Flensburg	IBM	Spirit AeroSystems	ICEYE
Diehl Group	Ceska zbrojovka	Saab	EP Barrus	AMZ Kutno
BAE Systems	Theon International	General Atomics	Northern Diver	CELTECH
NVL Group	German Naval Yards Holding	Amey	UK Ministry of Defence	Creotech
EuroDASS	RTX	KNDS	Rafael Advanced Defense Systems	General Electric
Secunet	Boomeranger Boats	UK Docks Marine Services	FN Group	Airbus
Kongsberg	Gocher Fahrzeugbau	Chemring Group	Tekever	Northrop Grumman
Saab	Patria Kand Oy	Safran	Anduril	Iveco Group
Iveco Group	Safran	General Dynamics	MSubs	Łukasiewicz Research Network
Elbit Systems	Defenture	Systems Engineering & Assessment	Edgar Brothers	Ford Motor Company
Eurofighter GmbH	Doll Fahrzeugbau	Computacenter	Bombardier	MIC Industries
Rafael Advanced Defense Systems	Leonardo	Cook Defence Systems	JFD	
L3Harris	Mercedes-Benz Group	Elbit Systems	Bohemia Interactive	
Atlas Elektronik		L3Harris	Serco	
Antonov Logistics SALIS		Supacat	Astroscale	
Heckler & Koch		Smiths Group	Rhea	
ESG Elektroniksystem- und Logistik		Centreline	Digital Concepts Engineering	
Panavia Aircraft		KBR	Cerca magnetics	
Metallwerk Elisenhütte		NP Aerospace	Fraunhofer	
Rohde & Schwarz		Fincantieri	Beam Global	
Deutsche Bahn		Atlas Elektronik	Zentraxa	

Figure A1: Value of SBIR/STTR awards by department, \$ billions, 2024 prices



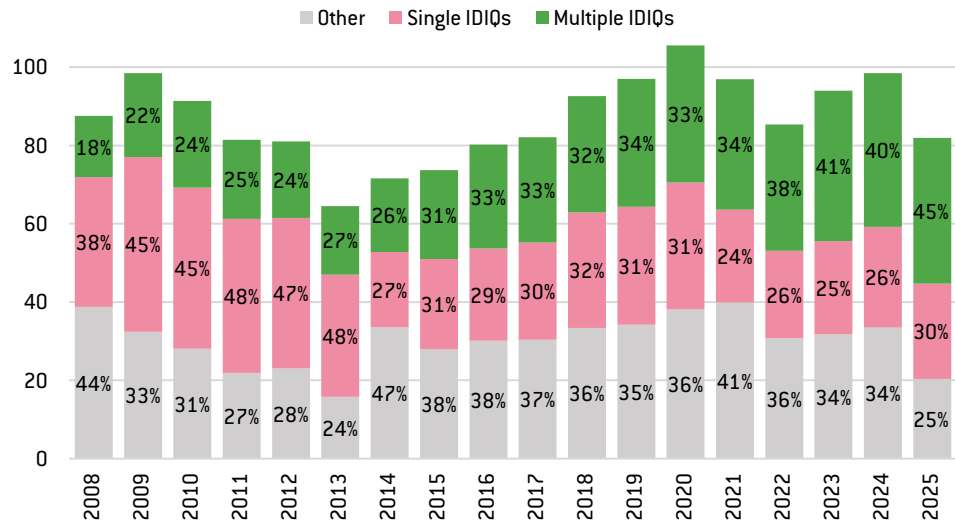
Source: Bruegel based on US GSA. Note: Department of Defense (DoD), General Services Administration (GSA), National Aeronautics and Space Administration (NASA), and Department of Homeland Security (DHS). 'Other' includes the Department of the Interior, the Department of Health and Human Services, the Department of Transportation and the remaining government departments.

Figure A2: Defence Innovation Unit (DIU), successful prototypes



Source: Bruegel based on DIU. Note: 'transitioned' refers to an Other Transaction Agreement (OTA) that led to a contract with the DoD or other government agency, while 'awarded, not transitioned' means that a proposal was awarded an OTA but this did not lead to a procurement contract.

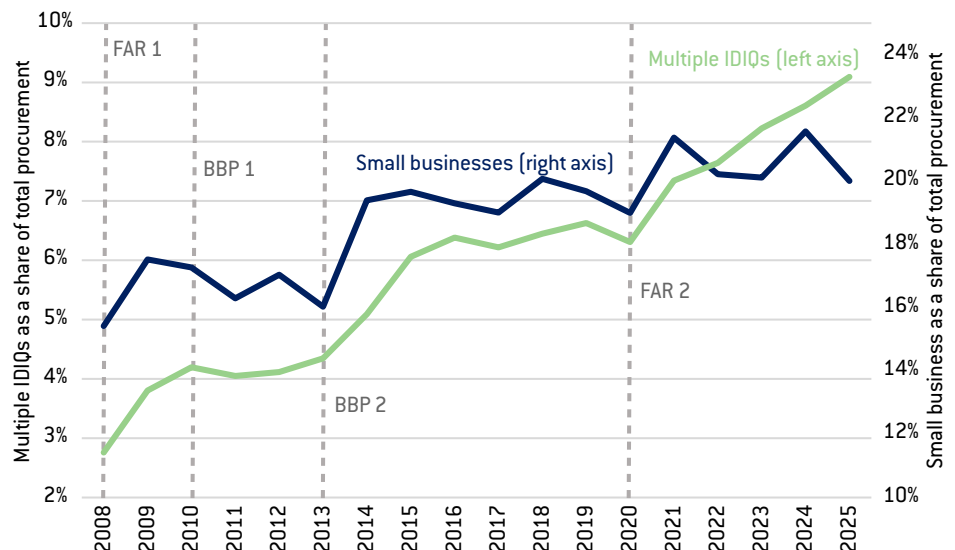
Figure A3: Value of contracts awarded to small businesses by contracting vehicle, \$ billions 2024 prices



Source: Bruegel based on US GSA. Note: 'multiple IDIQs' refers to contracts awarded through a multiple-award IDIQ vehicle, while 'single IDIQs' refers to contracts awarded through a single-award IDIQ contracting vehicle.

Figure A3 shows that since 2008, an increasing share of the value awarded to small businesses has come from contracts awarded under a multiple-award IDIQ. After the second BBP initiative in 2013 (see footnote 12), multiple-award IDIQs overtook single-award IDIQs in the share of small business awards. Since 2015, over 40 percent of the value of multiple award contracts has been awarded to small businesses, compared to under 20 percent of single award contracts. Since 2022, multiple-award IDIQs have become the single largest type of contracting vehicle for small businesses, accounting for 45 percent of DoD procurement from small businesses in 2025.

Figure A4: Multiple IDIQs and small businesses procurement as a share of total procurement



Source: Bruegel based on US GSA. Note: 'FAR 1' refers to a 2008 reform that made multiple-award IDIQs the preferred contracting method, and deterred the use of single-award IDIQs. 'FAR 2' refers to the 2020 reform that removed the requirement to evaluate price in multiple-award IDIQs. 'BBP 1' reformed acquisition processes by using contracting types to increase competition and 'BBP 2' focused on sustaining competition throughout the acquisition lifecycle and not just at the time of the award.

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